

Patterns and prediction: clustering, classification, leakage

- Örüntü ve tahmin: kümeleme, sınıflandırma ve veri sızıntısı
- BIB Week 9

Ad of the Week #9

- Skills Ledger: tick machine learning and model validation

The killer football

- BMJ published: "Cardiovascular mortality in Dutch men during the 1996 European football championship"
- 22 June 1996, Netherlands lost to France. Infarction and stroke deaths spiked.
- Proposed mechanism: alcohol and stress
- The arithmetic: a pattern like this appears by chance about once every 899 days
- There was a football match in the middle of it. There is always something in the middle of it.
- Witte DR et al., BMJ 2000;321:1552–1554

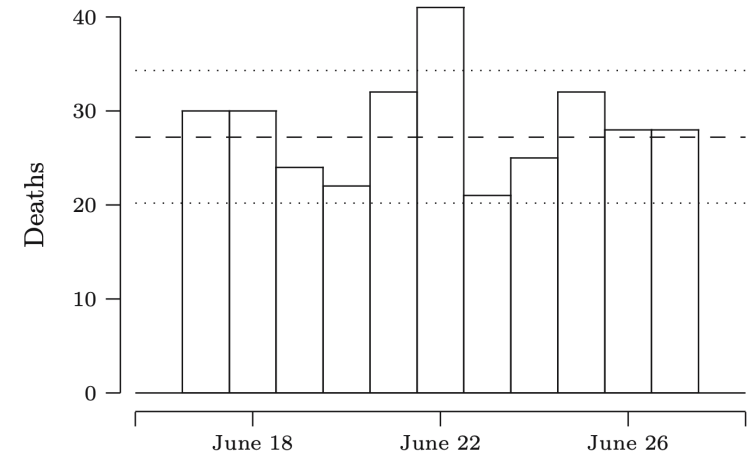
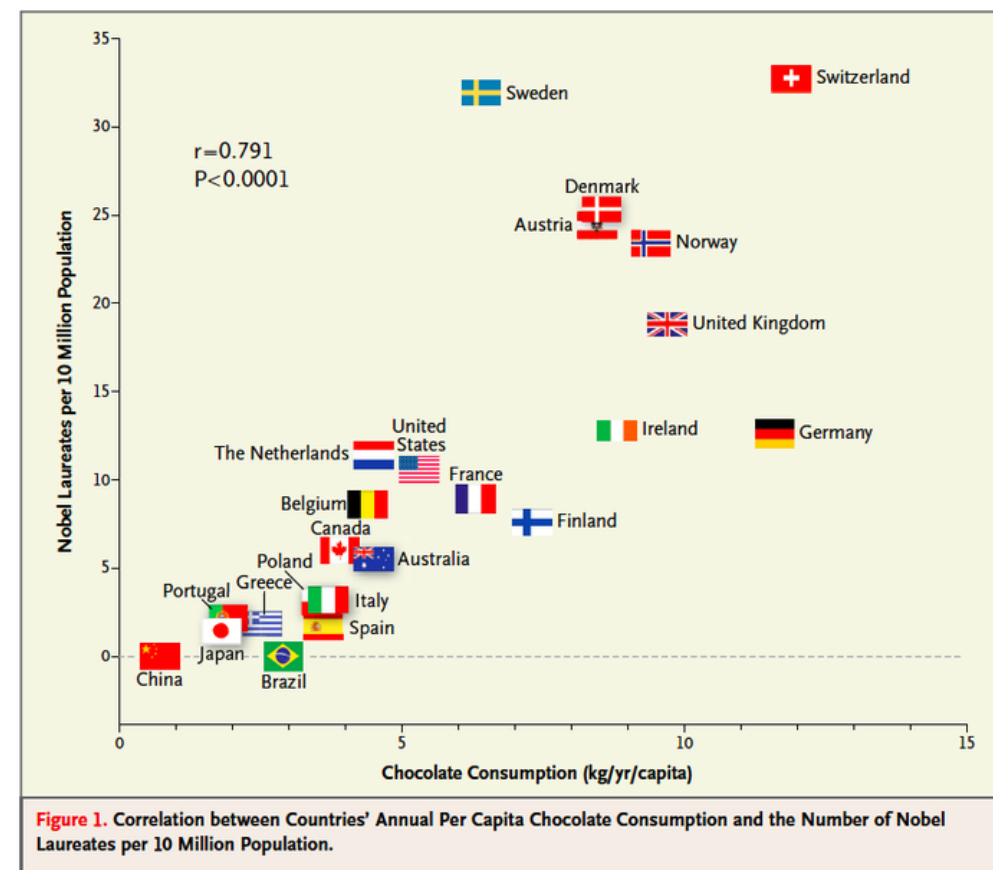


Fig. 1.2. Number of deaths from infarction or stroke in (part of) June 1996.

Deaths from infarction or stroke, Netherlands, June 1996 (Witte et al., BMJ 2000)

Chocolate and Nobel prizes

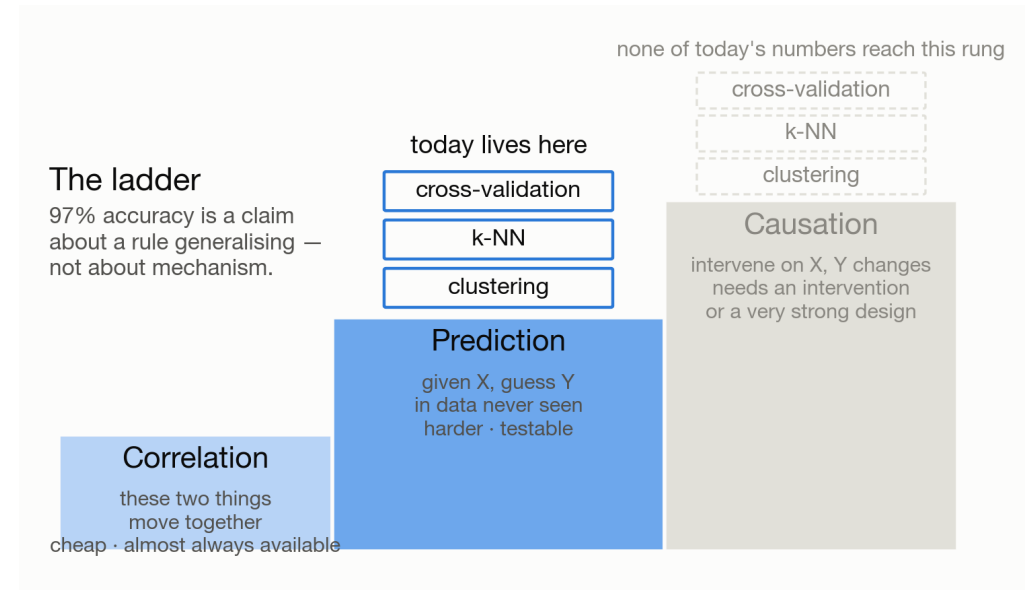
- Per-capita chocolate consumption vs Nobel laureates per capita, 23 countries
- $r = 0.791$, $p < 0.0001$
- Published in the New England Journal of Medicine — Messerli FH, NEJM 2012;367:1562–1564
- The correlation is real. The arithmetic is correct. The paper is a joke and was meant as one.
- Most spurious correlations are not jokes and nobody labels them



*Chocolate consumption vs Nobel laureates per capita, $r = 0.791$
(Messerli, NEJM 2012)*

The ladder, and where today lives

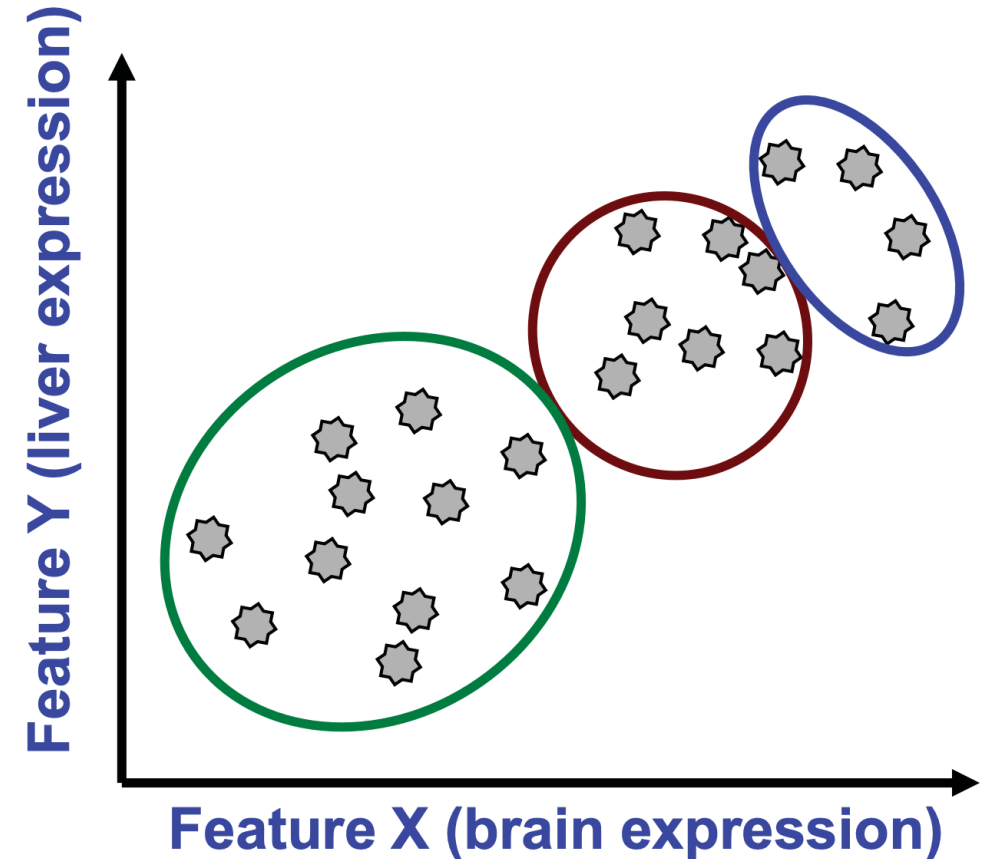
- Correlation — these two things move together. Cheap. Almost always available.
- Prediction — given X, I can guess Y in data I have not seen. Harder. Testable.
- Causation — if I intervene on X, Y changes. Requires an intervention or a very strong design.
- Everything in today's lecture lives on the middle rung.
- A model with 97% accuracy has made no claim whatsoever about mechanism
- Do not let anyone — including yourself — quietly promote a rung



Correlation, prediction, causation: today lives on the middle rung

Two different questions, not two toolboxes

- Unsupervised (denetimsiz öğrenme): are there groups in my samples I did not expect?
 - no labels; the output is structure, and structure is not validated by the method that found it
- Supervised (denetimli öğrenme): can I predict this label from expression?
 - labels exist; the output is a rule, and the rule is validated on data it never saw
- The difference is not which algorithm you type. It is what you already know.
- If you have labels and you cluster anyway, you have thrown away information on purpose — sometimes correctly



Clustering: structure without labels (CB Lecture_12 slide 6)

Variable types: encoding is a modelling decision

- Nominal (anma): categories with no order — tissue, batch, sex
- Ordinal (sıralı): ordered but unevenly spaced — tumour stage I/II/III, "low/medium/high"
- Continuous (sürekli): interval or ratio — expression, age, concentration
- Encoding stage as 1, 2, 3 asserts II–I = III–II — a biological claim made in a data-cleaning step
- One-hot encoding makes no such claim, and costs you two columns
- Distance-based methods (everything today) read your encoding literally

The same column, two encodings, two different claims (Euclidean distances shown)

encoded as 1 / 2 / 3

tumour stage	code
I	1
II	2
III	3

$$\begin{aligned}d(I, II) &= 1 & d(II, III) &= 1 \\d(I, III) &= 2\end{aligned}$$

✗ asserts stage III is twice as far from I as II is — a biological claim made in a data-cleaning step

one-hot encoded

tumour stage	is_I	is_II	is_III
I	1	0	0
II	0	1	0
III	0	0	1

$$\begin{aligned}d(I, II) &= d(II, III) \\&= d(I, III) = \sqrt{2}\end{aligned}$$

✓ every pair equally far: no ordering claimed — costs two extra columns

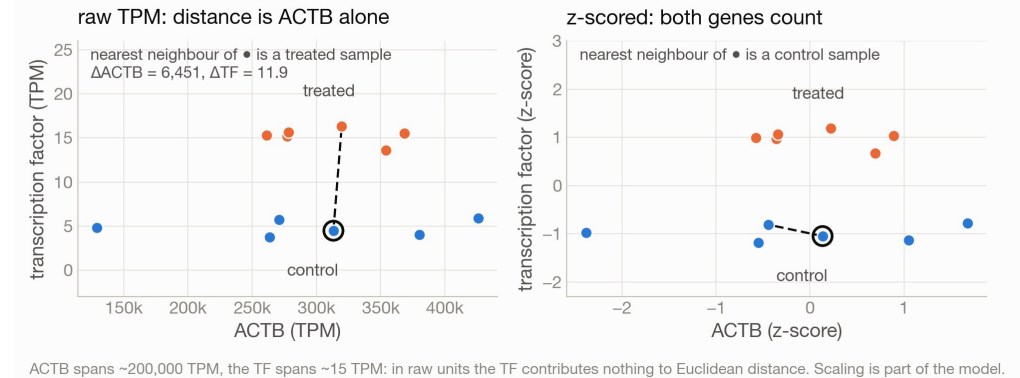
Distance-based methods — k-NN, k-means, hierarchical, PCA — read the encoding literally.

Tumour stage as 1/2/3 versus one-hot: the implied Euclidean distances

Distance depends entirely on scaling

- k-NN, hierarchical clustering, k-means, PCA — all of them are distance in disguise
- Two genes: gene A ranges 0–500,000 TPM, gene B ranges 0–20 TPM
- Euclidean distance between any two samples is essentially gene A alone. Gene B contributes nothing.
- Min-max, z-score, log transform — each is a different claim about what "far apart" means
- Scaling is not preprocessing. It is part of the model.
- Remember that sentence for minute 55, when scaling becomes a leakage pattern

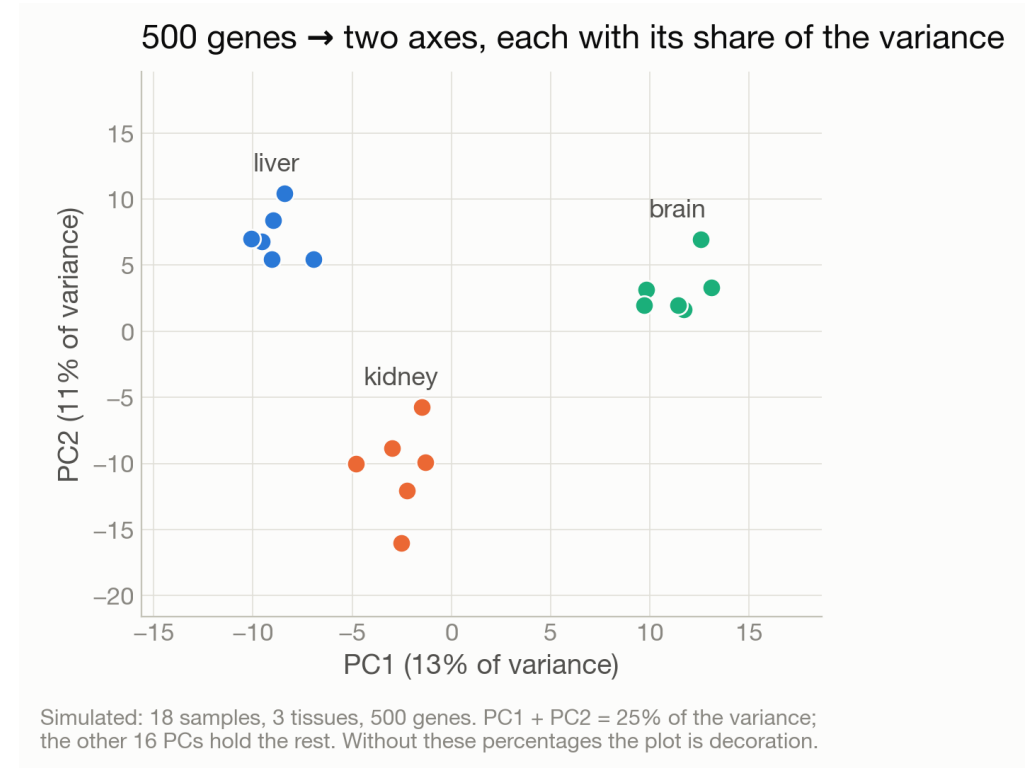
Same twelve samples, same two genes — the nearest neighbour depends on the scaling



ACTB (TPM) vs a low-expressed transcription factor: the nearest neighbour changes after z-scoring

PCA in words

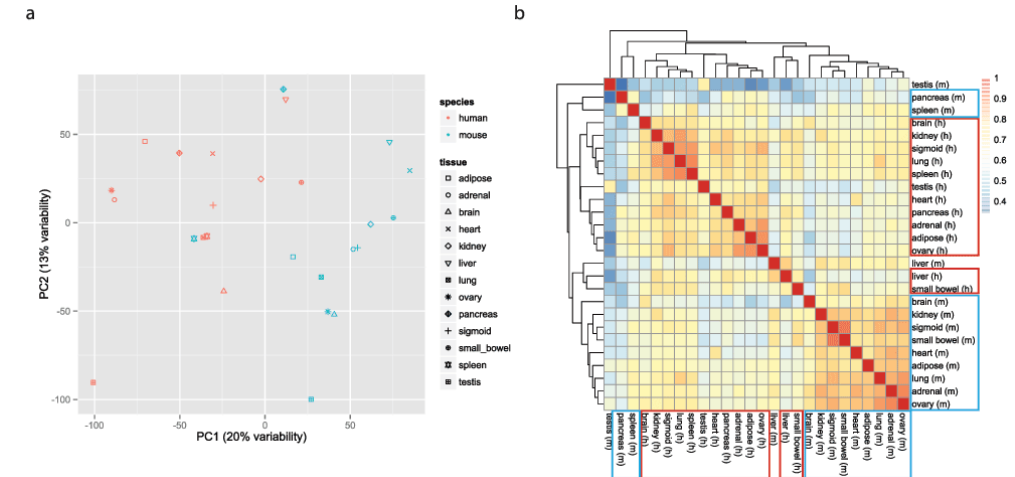
- Find the direction in which the samples vary most. Call it PC1. Repeat, at right angles.
- Each PC has a % variance explained — always report it; a PCA plot without it is decoration
- PCA is a rotation, not a discovery: it finds the biggest variance, not the most interesting one
- Useful because 20,000 genes are unplottable and the first 2–10 PCs usually carry the structure
- Novembre et al., Nature 2008;456:98 — PC1 and PC2 of European genotypes reproduce the map of Europe



PCA of 18 simulated tissue samples: PC1 (13%) and PC2 (11%) — the percentages are not optional

The figure that matters: PC1 is your sequencing date

- Twelve RNA-seq samples. PC1 explains 62% of the variance. It separates them perfectly — by processing day.
- Treatment is somewhere in PC4, if it is anywhere
- Leek et al., Nat Rev Genet 2010;11:733 — batch effects are widespread, critical, and routinely invisible
- Worst published case: Gilad & Mizrahi-Man, F1000Res 2015;4:121 — clustering was sequencing batch
- PCA is the first plot you make and the last one you show. It is a diagnostic, not a result.
- If condition is perfectly confounded with batch, no algorithm fixes it — Week 6, verbatim



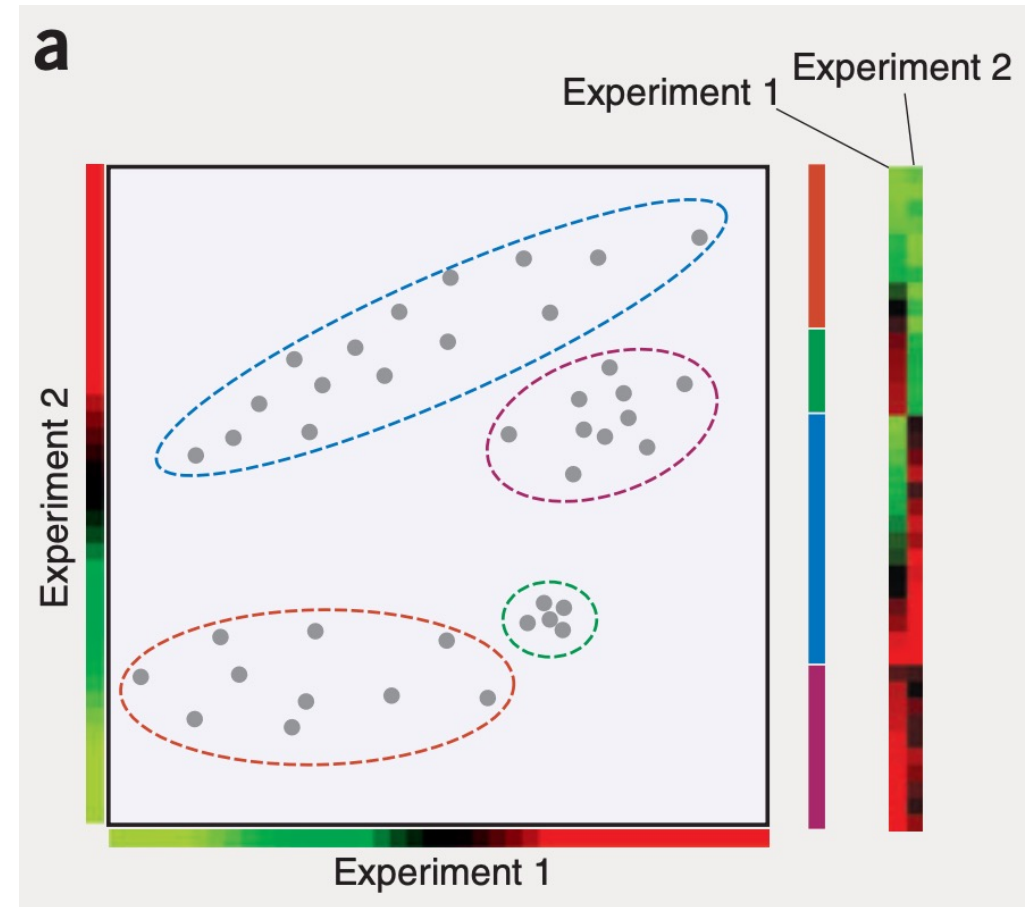
Gilad & Mizrahi-Man, F1000Research 2015;4:121, Fig. 2 (CC-BY): mouse and human ENCODE samples cluster by species — which was the sequencing batch

Two minutes of health warning on UMAP

- UMAP and t-SNE produce a picture, not a metric
- Distances between clusters mean nothing. Cluster sizes mean nothing. Empty space means nothing.
- The same data with a different `n_neighbors` gives a different picture, and both are "correct"
- Chari & Pachter, PLoS Comput Biol 2023;19:e1011288 — "The specious art of single-cell genomics"
- Legitimate: showing a structure you found by other means. Illegitimate: measuring anything off it.
- You will meet UMAP again in Week 10 and I will say this again

Clustering, taught honestly

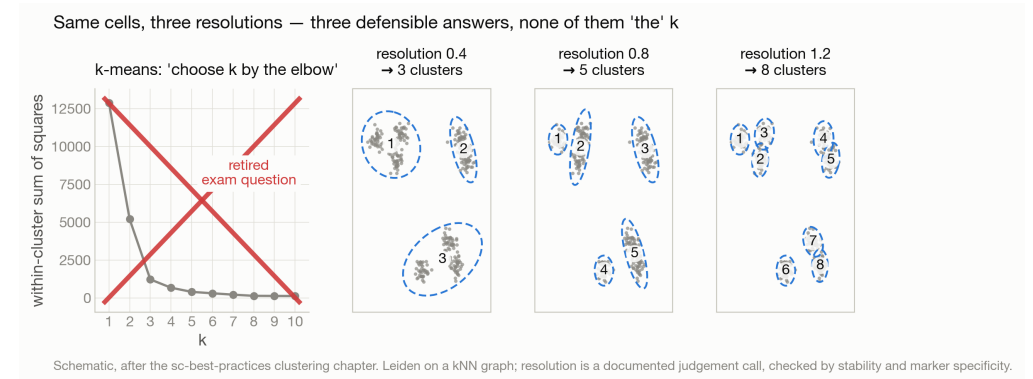
- Hierarchical (hiyerarşik kümeleme): build a tree bottom-up
 - the linkage rule (single/complete/average/Ward) changes the tree, so state it
- k-means: pick k, assign to nearest centroid, move centroids, repeat until nothing moves
- Both are legitimate and standard for bulk expression data, and both are the historical default
- Neither validates itself. The clusters exist because you asked for clusters.
- D'haeseleer, Nat Biotechnol 2005;23:1499 — "How does gene expression clustering work?"



Two experiments, 40 genes: clusters in expression space (D'haeseleer 2005)

What single-cell does instead

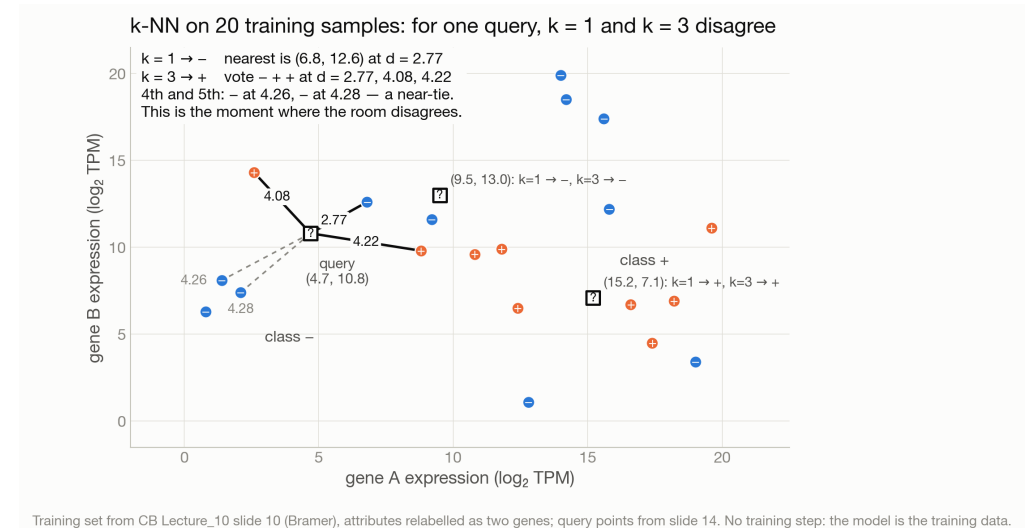
- Single-cell data is not clustered with k-means, and this is not a stylistic preference
- The field uses Leiden on a k-nearest-neighbour graph over PCA or a learned latent space
 - Traag, Waltman & van Eck, Sci Rep 2019;9:5233
- The tunable is resolution (çözünürlük), not k — higher resolution splits, lower resolution merges
- No elbow criterion for resolution, and no objective k. It is a documented judgement call.
- Checked by cluster stability and marker specificity, then written down
- This course's question pool asked for the elbow method. That question is retired.



The retired elbow plot versus the same cells at three resolutions (3, 5 and 8 clusters)

k-NN, by hand, on eight points

- Plot the eight labelled training points on the board. Add one unlabelled query point.
- $k = 1$: the query takes the label of its single nearest neighbour
- $k = 3$: majority vote of the three nearest
- Draw the decision boundary. Watch it change as k changes.
- Small $k \rightarrow$ jagged boundary, fits noise. Large $k \rightarrow$ smooth boundary, ignores small real groups.
- There is no training step. The model is the training data.



k-NN on the 20 training points: for query (4.7, 10.8), $k = 1$ says $-$ and $k = 3$ says $+$

The confusion matrix: precision and recall

- Four cells: TP, FP, FN, TN. Everything else is arithmetic on those four.
- Precision (kesinlik) = $TP / (TP + FP)$ — of the ones I flagged, how many were real?
- Recall / sensitivity (duyarlılık) = $TP / (TP + FN)$ — of the real ones, how many did I find?
- They trade off. A model that flags everything has recall 1.0 and useless precision.
- Which one you care about is a clinical or engineering decision, not a statistical one
- Screening test → recall. Confirmatory test → precision.

Correct classification	Classified as	
	+	—
+	true positives	false negatives
—	false positives	true negatives

The four cells: TP, FN, FP, TN

The base rate will humiliate you

- Cohort prevalence 3%. A classifier that predicts "healthy" for everyone is 97% accurate.
- Recall = 0. Precision = undefined. Accuracy = 97%. Published.
- Accuracy (doğruluk) is only interpretable against the base rate (taban oran)
- Class imbalance (sınıf dengesizliği) is the normal condition in biology, not an edge case
- AUROC summarises the precision/recall trade-off across all thresholds — named, not derived
- With heavy imbalance, prefer precision–recall curves; AUROC looks flattering on rare classes

	predicted: disease	predicted: healthy	
actual: disease	TP 0	FN 30	30 patients (3%)
actual: healthy	FP 0	TN 970	970 patients (97%)

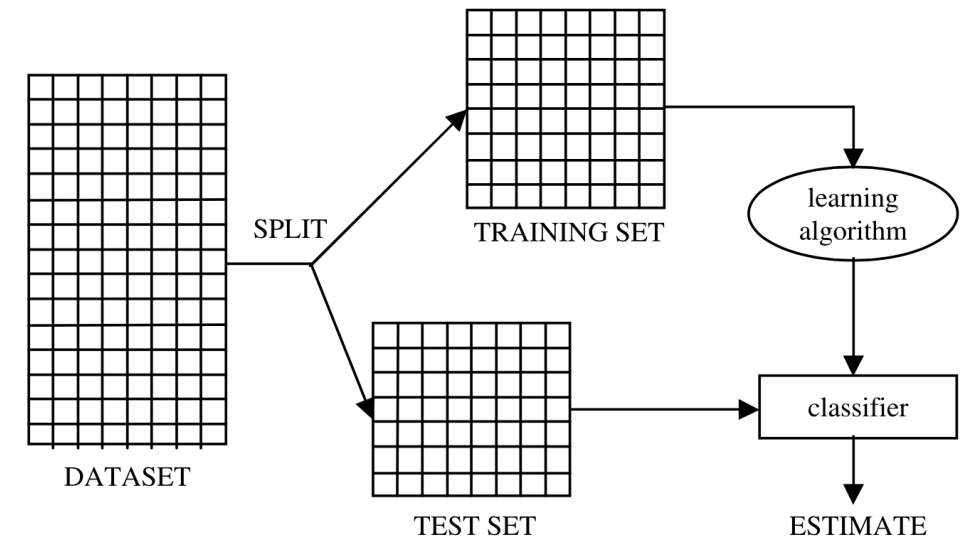
Accuracy = $(0 + 970) / 1000 = 97\%$

Recall = $0 / 30 = 0$ Precision = $0 / 0 = \text{undefined}$ The model: predict 'healthy'. Always.

Predict 'healthy' for everyone: 97% accurate, recall 0

Train, test, and the only honest question

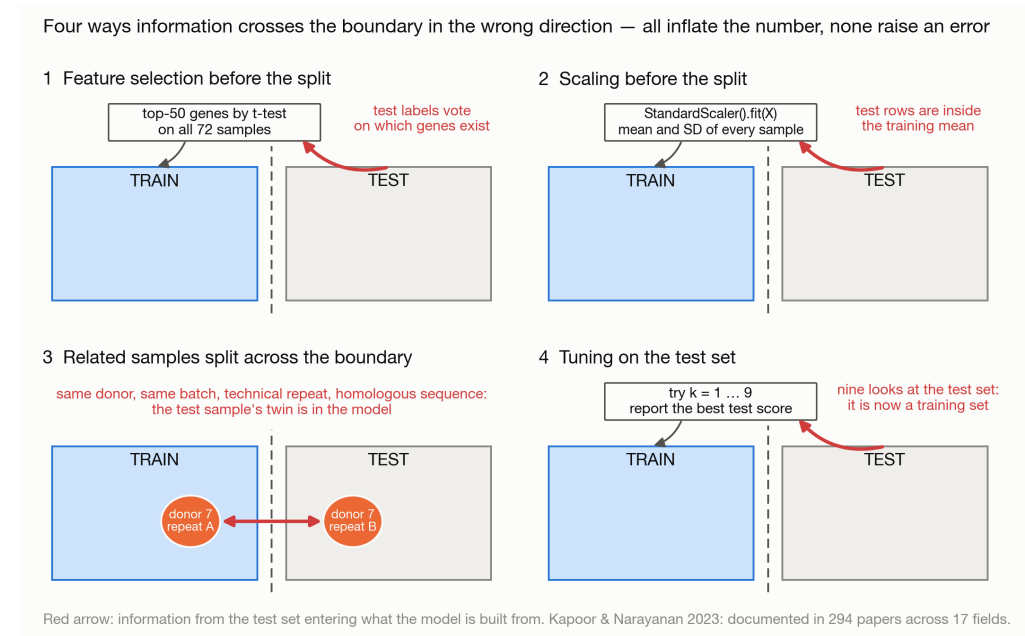
- Train/test split (eğitim/test ayrımı): fit on one part, measure on a part the model never saw
- k-fold cross-validation (çapraz doğrulama): split into k parts, hold each out in turn, average the k scores
 - preferred when n is small, which in biology is always
- Accuracy on data the model was fitted to is not an estimate of anything. It is a memory test.
- The only question worth asking: on what data was this number measured, and had the model seen it?
- Everything in the next twelve minutes is a way of answering "yes" to that question by accident



Train/test split (CB Lecture_11 slide 6)

Four ways the test set stops being a test set

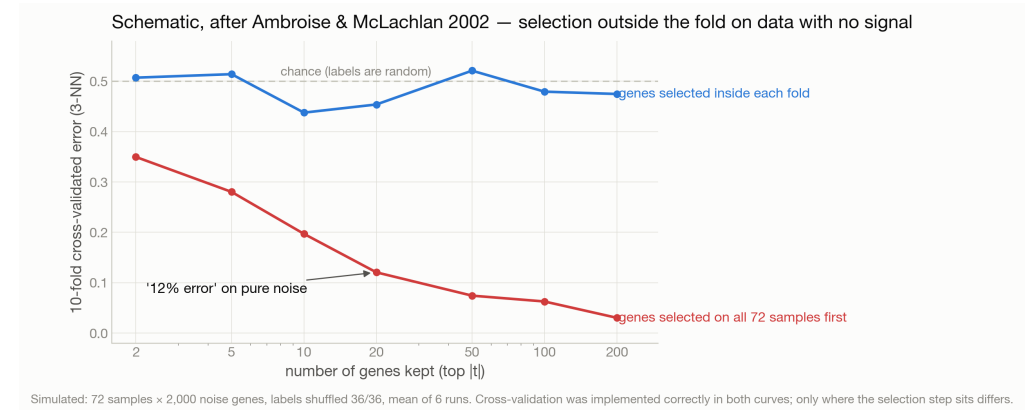
- 1. Feature selection on all the data before splitting. You chose the genes using the test labels.
- 2. Scaling or normalising before splitting. Test-set mean and variance are in the training data.
- 3. Related samples split across the boundary. Same donor, same batch, technical repeats.
- 4. Tuning on the test set. You tried nine values of k and reported the best one.
- All four inflate the reported number. None of them raise an error.
- Kapoor & Narayanan, Patterns 2023;4:100804 — leakage documented in 294 papers across 17 scientific fields



Four ways information crosses from test to train

The evidence, on data exactly like yours

- Ambroise & McLachlan, PNAS 2002;99:6562 — selection bias in gene extraction from microarray data
- They took expression data with no real signal, selected genes on the full set, then cross-validated
- Reported error rate: near zero. On noise.
- With gene selection done correctly inside each fold, the error rate returned to chance
- This is not a subtle statistical point. It is the difference between a discovery and nothing.
- The data was ALL/AML leukaemia expression — the same dataset you will use in forty minutes



Schematic, after Ambroise & McLachlan 2002: gene selection outside the fold reports near-zero error on pure noise

The sentence that anchors your lab

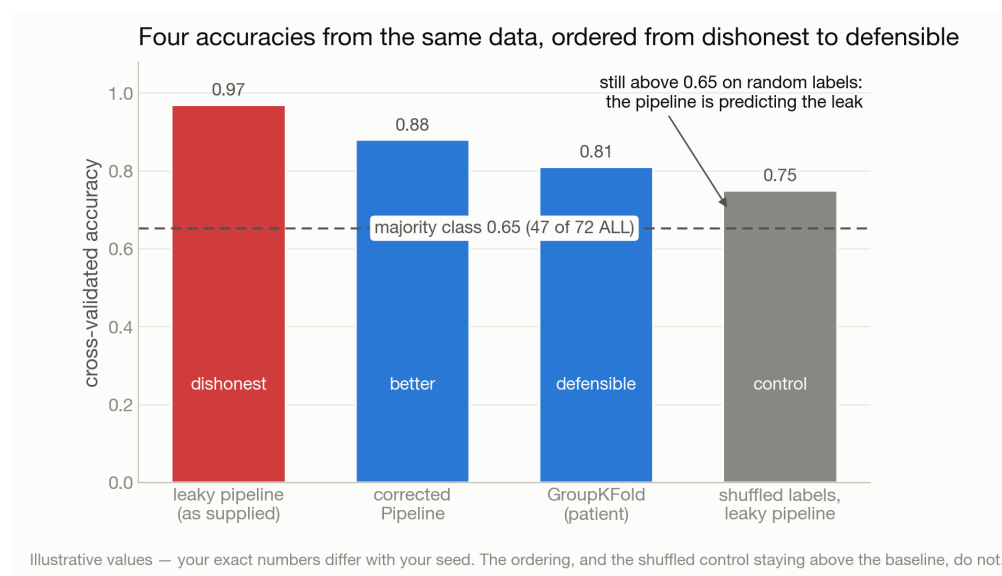
- Cross-validation cannot detect pattern 3.
- Every fold contains the same contamination. The average of ten contaminated folds is a contaminated number.
- Twelve of your samples appear twice as technical repeats; random k-fold puts a twin in the training set
- k-NN then classifies the test sample by finding... its own twin.
- The fix is not a better metric. It is `GroupKFold(groups=patient_id)` — one supplied line.
- The unit of independence (bağımsızlık birimi) is the patient, not the sample



KFold puts donor 7's twin in the model whenever its partner is tested; GroupKFold keeps them together

The control that cannot be argued with

- Shuffle the labels. Destroy every real association. Re-run the identical pipeline.
- An honest pipeline returns chance — about 0.65 here, the majority-class rate
- A leaking pipeline still returns ~ 0.75 on shuffled labels, because it is fitting the leak, not the biology
- This is the negative control from Week 6, applied to a model instead of to a p-value
- It costs one line and it is the strongest evidence you can produce about your own work
- Nobody can argue with a number produced from randomised labels



Four accuracies from the same data, majority-class baseline 0.65 dashed

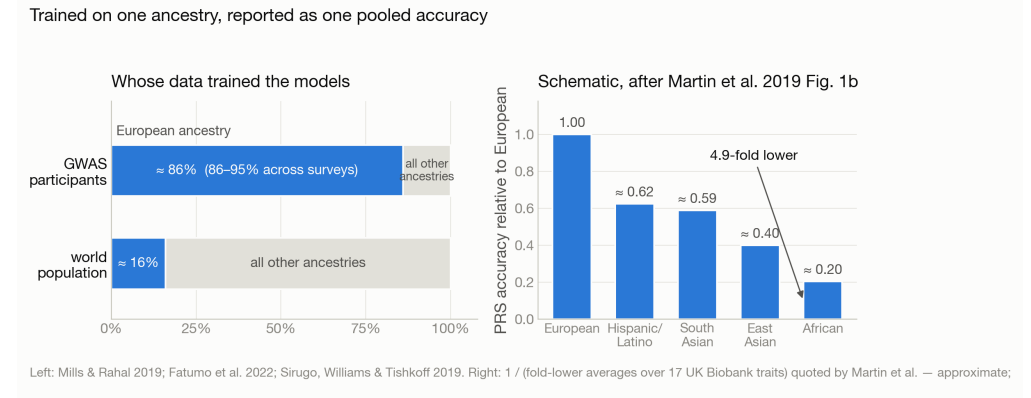
Baseline discipline

- Always report the trivial baseline before the model. Majority class. One feature. A coin.
- Golub ALL/AML: 47 of 72 samples are ALL, so the majority-class baseline is 0.653
- A model at 0.71 is not "71% accurate". It is six points of information over guessing.
- "A benchmark without a strong simple baseline is not a benchmark."
- This sentence returns in Week 11 as the entire argument of the lecture

A benchmark without a strong simple baseline is not a benchmark.

Ethics beat: the accuracy you did not break down

- A classifier trained on one ancestry group, reported as a single pooled accuracy number
- Polygenic scores transfer poorly across ancestries (Martin et al., Nat Genet 2019;51:584)
- Roughly 86–95% of GWAS participants have been of European ancestry
- Türkiye's population is not well represented in any of the major reference panels
- A pooled number hides the subgroup where the model fails, and the subgroup is usually the one with least data



Left: European-ancestry share of GWAS participants (~86%) vs world population (~16%). Right: schematic, after Martin et al. 2019 — PRS accuracy relative to European ancestry, African ≈ 0.20

The rule you write down, and what the lab does

- Report performance per subgroup, or state that you did not. There is no third option.
- "We did not have enough samples to stratify" is an acceptable sentence. Silence is not.
- Your seed's data has a cohort column with two groups of different class balance
- At home: accuracy per cohort, find where it collapses, one sentence on what the pooled number hid

Per-subgroup report: the pooled 0.90 hides cohort B (15% of the data, 7 of 11 right)

subset	n	class balance	majority baseline	accuracy	over baseline
pooled (all 72)	72	47/72 ALL (65%)	0.65	0.90	+0.25
cohort A	61	42/61 ALL (69%)	0.69	0.95	+0.26
cohort B	11	5/11 ALL (45%)	0.55	0.64	+0.09

Illustrative counts. The AUDIT section of Design Notebook 2 produces exactly this shape, on your seed's `cohort` column.

Pooled 0.90 versus 0.95 / 0.64 per cohort; cohort B highlighted